Advanced Mathematical Economics

First Order Difference Equations (Numerical Problem)

Dr. Hafiz Muhammad Qasim

Assistant Prof. Department of Economics

Akhuwat University

Hmq.economist@gmail.com

Outlines

- Summery of Previous Lecture
- Numerical Example

Summary of Previous Lecture

- In previous lecture we have derive the following general and definite solutions of first order difference equations:
- Satndard form of first order difference equation
- $\mathbf{Y_{t+1} + aY_{\mathit{t}} = C}$
- Non-Homogeneous Case if C≠0
- $\mathbf{Y_{t+1} + aY_{\mathit{t}} = C}$
- Homogeneous Case if C=0
- $\mathbf{Y_{t+1} + aY_{\mathit{t}} = 0}$

Summary of Previous Lecture

- **The objective of difference equations is to find the time path:**
- $Y_t = Y_c + Y_p \dots \dots . . (4)$
- **Particular Solution or equilibrium solution (Yp)**
- $Y_p = \frac{c}{(1+a\ \ )} \dots \dots \dots (8)$
- **Complementary faction (Yc)**
- $Y_c = A(-a)^t \dots \dots \dots (10)$

Summary of Previous Lecture

- **Obtaining General Solution; if a≠0**
- $Y_t = \mathbf{A}(-a)^t + \frac{c}{(1+a\ \ )} \ldots\ldots(11)$
- **Definite Solution; a≠-1**
- **A**$= Y_0 - \frac{c}{(1+a\ \ )}$**.........(12) if t=0**
- $Y_t = \{Y_0 - \frac{c}{(1+a\ \ )}\}(-a)^t + \frac{c}{(1+a\ \ )} \ldots\ldots(13)$

Summary of Previous Lecture

- **General Solution ; a= -1**
- $Y_t = \mathbf{A} + Ct \ldots..(14)$
- **Definite Solution; if a= -1**
- $\mathbf{A} = Y_0 \ldots\ldots(15) assuming\ t = 0$
- $Y_t = Y_0 + Ct \ldots..(16)$

Example 04: Page 550

- Q: Solve the following first-order difference equation
- $\mathbf{Y_{t+1} - 5Y_{\mathit{t}} = 1; \ (Y_0 = \frac{7}{4})}$
- $\mathbf{Y_{t+1} + aY_{\mathit{t}} = C}$ **(By comparing with standard form)**
- **a=-5 & c=1**
- $\mathbf{Y_{\mathit{c}} = A}(-a)^t \ldots \ldots \ldots (\mathbf{10})$
- $\mathbf{Y_{\mathit{c}} = A}(-(-5))^t$ **by substituting the value of** $(-\mathbf{a})$
- $\mathbf{Y_{\mathit{c}} = A}(5)^t$

For Particular Solution

- $Y_p = \frac{c}{(1+a\;\;)}$ ....(8)
- $Y_p = \frac{1}{(1+-5\;\;)}$ (by substituting c=1, and a=-5)
- $Y_p = -\frac{1}{4}$

For General Solution

- $Y_t = Y_c + Y_p \ldots \ldots .(4)$
- $Y_t = A(-a)^t + \frac{c}{(1+a\quad)} \ldots .(11)$
- $Y_t = A(5)^t - \frac{1}{4}$

For Definite Solution

- Assume t=0
- $Y_t = A(5)^t - \frac{1}{4}$
- $Y_0 = A(5)^0 - \frac{1}{4}$ (by putting t=0)
- $(Y_0 = \frac{7}{4})$; (as we know, Y0 given)
- $\frac{7}{4} = A(5)^0 - \frac{1}{4}$; (solving for A)
- $\frac{7}{4} = A(1) - \frac{1}{4}$; becase $(5)^0$=1
- $A = \frac{7}{4} + \frac{1}{4} = (\frac{7+1}{4}) = \frac{8}{4}$=2
- A=2

Definite Solution

- $Y_t = \{Y_0 - \frac{c}{(1+a\ \ )}\}(-a)^t + \frac{c}{(1+a\ \ )} \ldots\ldots$ **(13)**
- $Y_t = 2(5)^t + \frac{1}{4}$; $(Definite\ Solution)$

Dynamic Stability

- The dynamic stability of difference equations depends absolute value of exponential growth component; $b^t = (-a)^t$

- If $|b| > 1$ In this case, the absolute value of b is greater than 1, which means that the exponential term b^t grows rapidly as t increases. This leads to unbounded growth of Y_t over time, indicating dynamic instability.

$|b| < 1$

- If $|b| < 1$: Conversely, if the absolute value of b is less than 1, the exponential term b^t decreases rapidly as t increases. In this scenario, $Y_t \rightarrow 0$ approaches zero over time, indicating dynamic stability.

$|b \quad| = 1$

- If $|\mathrm{b} \quad| = 1$: When the absolute value of b equals 1, the exponential term b^{t} remains constant over time. In this case, the behavior of Y_t depends on the constant term C. If C≠0 , Y_t does not converge and exhibits dynamic instability. If C=0, the sequence remains constant over time, indicating dynamic stability.

- **Thank You**